ARQUITECTURAS E INFRAESTRUCTURAS PARA **INTELIGENCIA ARTIFICIAL**

Unit 3

Cloud Computing

Department: d:tic

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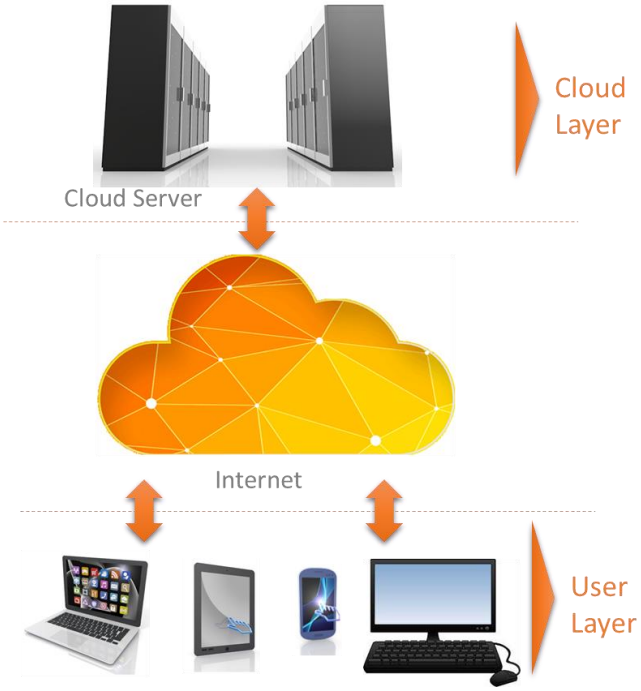
1. Cloud Computing Paradigm

The NIST Definition of Cloud Computing (2011)

Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.

NIST: National Institute of Standard and Technology
https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-145.pdf

1. Cloud Computing Paradigm

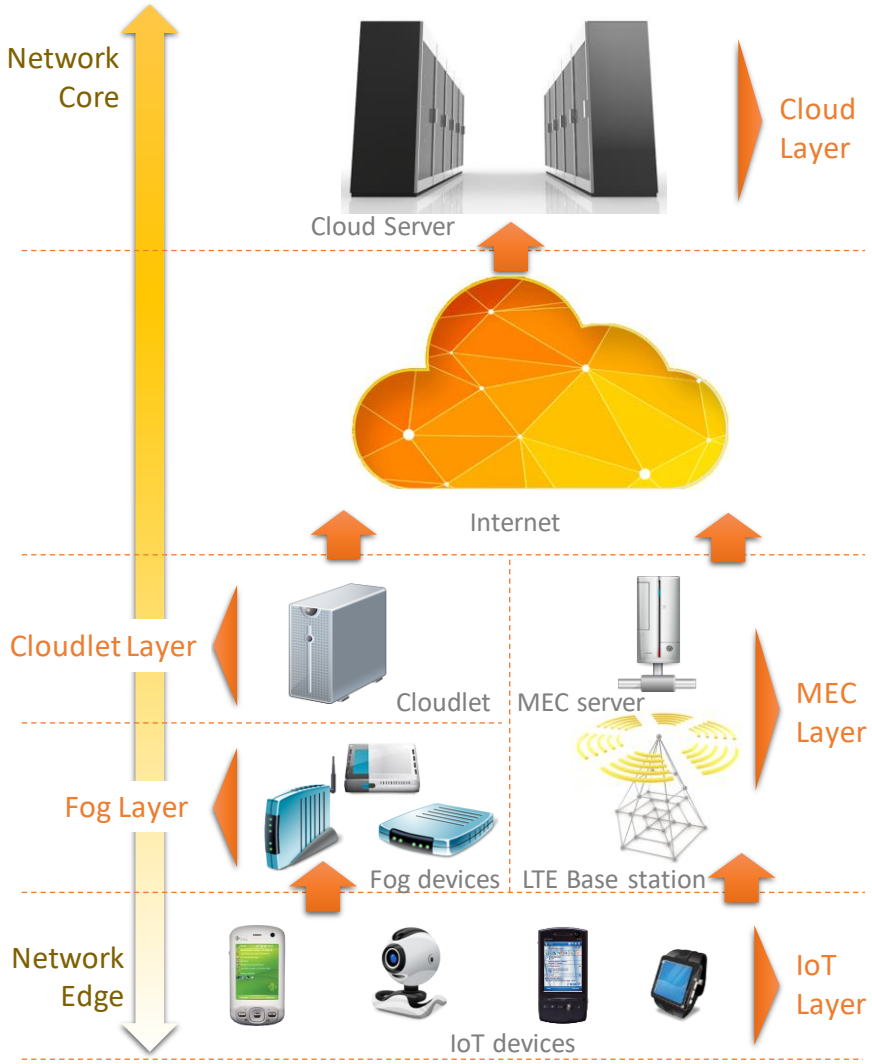


1. Cloud Computing Paradigm

- IT Service Externalization
  - Externalization of processing
  - Externalization of storage
  - Externalization of services
- *Mobile Cloud Computing* Paradigm (MCC)
  - Externalize computatioin of mobile and other IoT devices

1. Cloud Computing Paradigm

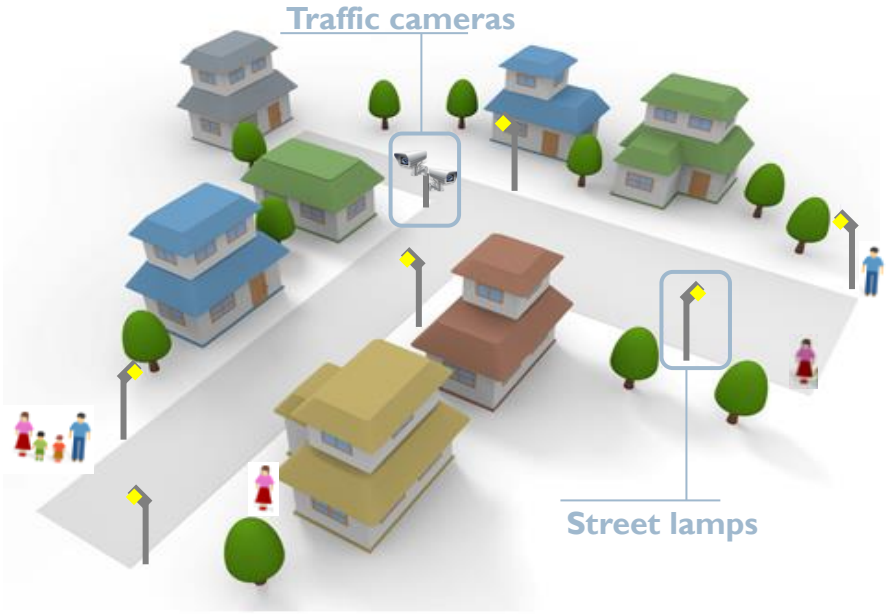
- Multilevel Cloud architecture
- Edge Computing concept

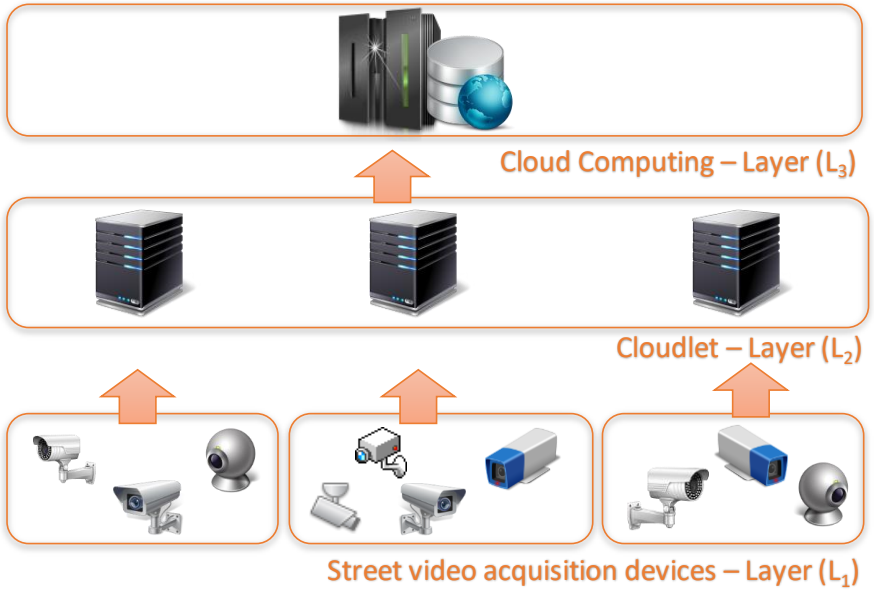


- Mora H. et al., (2019) Multilayer architecture model for mobile cloud computing paradigm, Complexity, 1-13. https://doi.org/10.1155/2019/3951495
- Kong et al., (2022) Edge Computing for Internet of Everything: A Survey, IEEE Internet of Things Journal, Volume: 9, Issue: 23 https://doi.org/10.1109/JIOT.2022.3200431

1. Cloud Computing Paradigm

- Example in Smart Cities

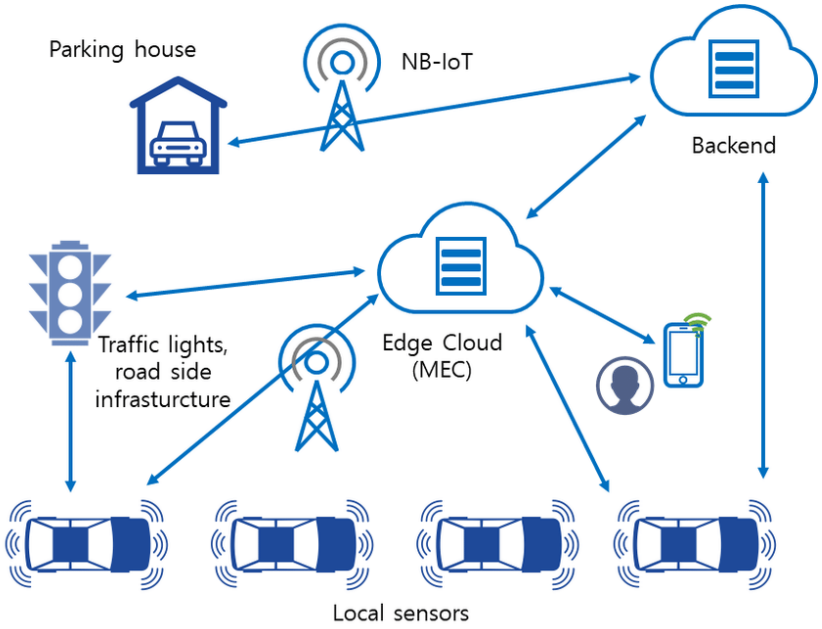




- Mora H. et al., (2019) Distributed Architectures for Intensive Urban Computing: A Case Study on Smart Lighting for Sustainable Cities. IEEE Access, Vol. 7. https://doi.org/10.1109/ACCESS.2019.2914613

1. Cloud Computing Paradigm

- Example in autonomous car



- Yeruva V. (2021) Edge Computing's Applications In Autonomous Driving And Business At Large https://www.forbes.com/sites/forbestechcouncil/2021/09/27/edge-computings-applications-in-autonomous-driving-and-business-at-large/?sh=5d470bd996aa
- Cui et al., (2020) Offloading Autonomous Driving Services via Edge Computing, IEEE Internet of Things Journal (Volume: 7, Issue: 10). https://doi.org/10.1109/JIOT.2020.3001218

1. Cloud Computing Paradigm

- Cloud paradigm advantages
  - Abstraction of resources
  - Scalability
  - Pay-per-use services (as-a-service)
  - Ubiquitous acess
  - Security
- Inconvenientes del paradigma Cloud
  - Shared infrastructure / performance issues
  - Technology dependence
  - Delocalization and loss of control of data
  - Higher risk of data breaches and leaks
  - Equalization of competitive advantage

2. Cloud Deployment Models

- Modelos de despliegue Cloud
  - Private Cloud
  - Public Cloud
  - Hybrid Cloud
  - Esquema Multicloud
  - Modelo Clouds Federadas

2. Cloud Deployment Models

- Public Cloud
  - The cloud infrastructure is provisioned for open use by the general public. It may be owned, managed, and operated by a business, academic, or government organization, or a combination of them. It exists on the premises of the cloud provider.

2. Cloud Deployment Models

- Private Cloud
  - The cloud infrastructure is provisioned for the exclusive use of a single organization comprising multiple consumers (for example, business units). It may be owned, managed, and operated by the organization, a third party, or a combination of both, and it may exist either on-premises or off-premises.

2. Cloud Deployment Models

- Hybrid Cloud
  - The cloud infrastructure is a composition of two or more distinct cloud infrastructures (private or public) that remain unique entities but are bound together by technology that enables data and application portability (for example, cloud bursting for load balancing between clouds).

2. Cloud Deployment Models

- Comparison of Cloud deployment model: **on-premise** vs **provider**.
  - Customization and Scalability
    - On-Premise: allows exclusive and customized use of data and hardware, but has limited capacity.
    - Provider: there is no absolute control over the data, but scalability is potentially unlimited.
  - Security
    - On-premise: ensures that the people who have access to the data have been previously authorized by the company.
    - Provider: the company entrusts its data to a third party.
  - Costs:
    - On-premise: Requires a high initial investment.
    - Provider: Pay-as-you-go model from day one.

2. Cloud Deployment Models

- MultiCloud Deployment Model
  - Multicloud describes the use of more than one cloud service provider
  - An orchestrator is required to manage cloud resources from different providers.
  - Advantages:
    - Reduces technological dependency.
    - Improves flexibility in the use of resources.
  - Disadvantages:
    - Higher cost and maintenance complexity.
    - Greater security risk.



2. Cloud Deployment Models

Federated Cloud Deployment



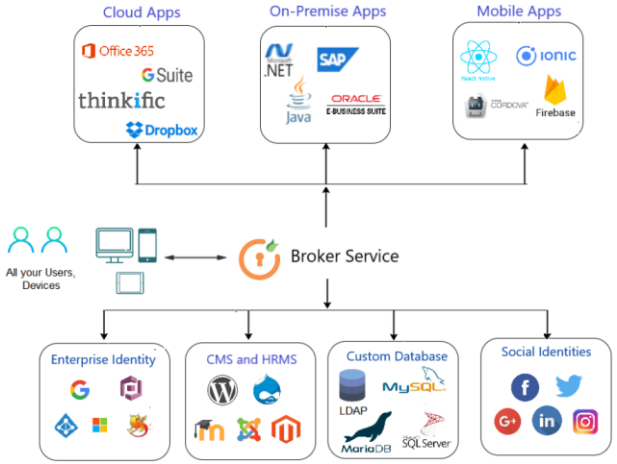
- A Federated Cloud is an ecosystem in which multiple cloud providers share services transparently to the user.
- Advantages:
  - Improves efficiency and reduces costs in the use of resources.
  - Increases reliability and scalability.
- Disadvantages:
  - Greater maintenance complexity.
  - Higher security risk.
  - Increased interoperability requirements.
- Gaia-X Project: The European Federated Cloud
  - https://www.gaia-x.eu/
  - https://www.gaiax.es/

gaia-x

- Kar B. Et al. (2023) Offloading using Traditional Optimization and Machine Learning in Federated Cloud-Edge-Fog Systems: A Survey. IEEE Communications Surveys & Tutorials. https://doi.org/10.1109/COMST.2023.3239579
- Li Y. et al. (2022) Federated Clouds for Efficient Multitasking in Distributed Artificial Intelligence Applications. IEEE Transactions on Cloud Computing. https://doi.org/10.1109/TCC.2022.3184157

2. Cloud Deployment Models

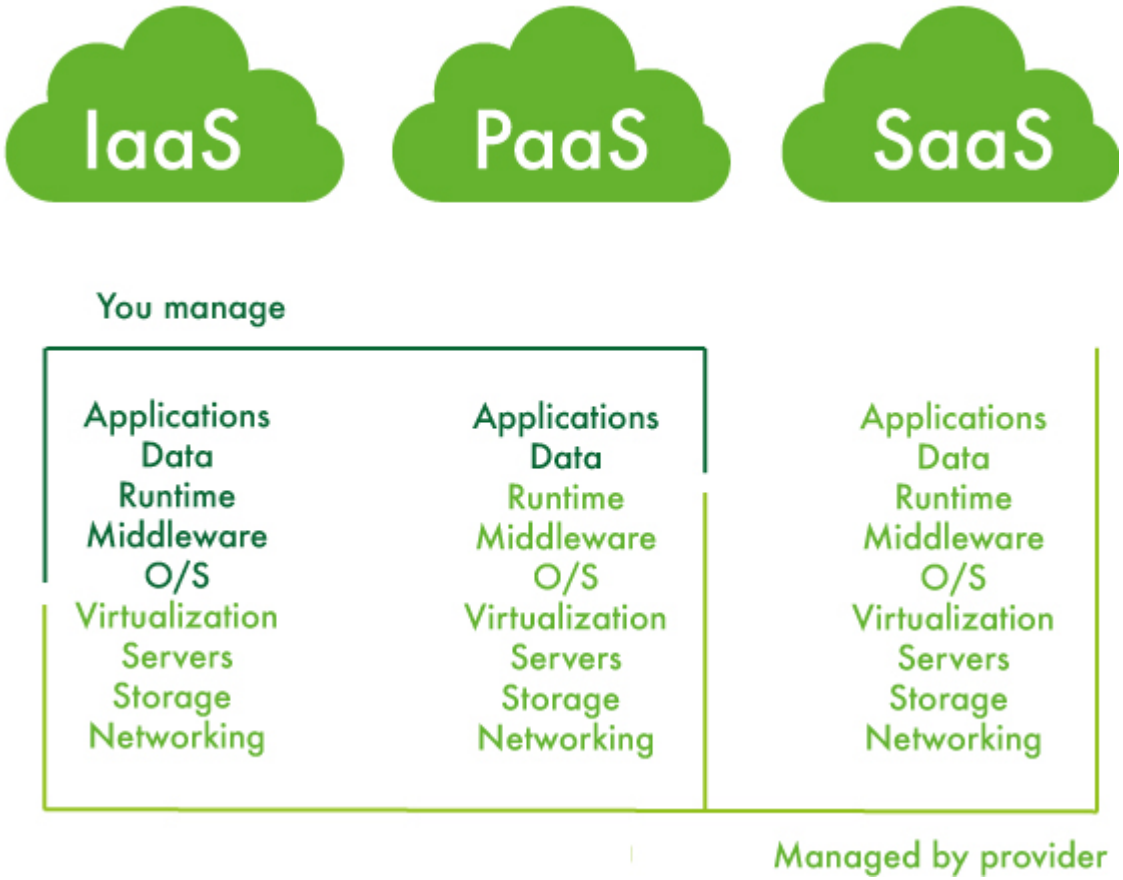
- Federated Cloud Deployment
  - A Broker Cloud service is necesary
    - Copilot said: Negotiate terms and conditions with the different Cloud providers
    - SLA: Service level agreements



- C. Muralidharan, R. Anitha, (2022) Trusted cloud broker for estimating the reputation of cloud providers in federated cloud environment, Concurrency and Computation, Vol. 34, Issue 11, https://doi.org/10.1002/cpe.6537

3. Service Models

- Service Models



3. Service Models

IaaS: Infrastructure as a Service

- The provided capability consists of delivering processing, storage, networking, and other fundamental computing resources on which software can be deployed and run.
- The consumer does not manage or control the underlying cloud infrastructure but has control over operating systems, storage, and deployed applications, and possibly limited control over certain networking components.
- Advantages:
  - Improved IT infrastructure management.
  - Cost savings and greater efficiency in the use of IT resources.
- Disadvantages:
  - Increased dependency on the cloud provider.
- Examples: Microsoft Azure, Amazon EC2

3. Service Models

PaaS: Platform as a Service

- The provided capability consists of deploying applications created or acquired by the consumer onto the cloud infrastructure, using programming languages, libraries, services, and tools supported by the provider.
- The consumer does not manage or control the underlying cloud infrastructure, including the network, servers, operating systems, or storage, but has control over the deployed applications and possibly over the configuration settings of the application hosting environment.
- Advantages:
  - Enables outsourcing of software development processes.
  - Improves productivity and reduces development costs, as there is no need to maintain the platform, databases, servers.
- Disadvantages:
  - Possible dependency on the provider in strategic aspects.
- Examples:
  - Velneo (http://velneo.es/). Desarrollo de aplicaciones empresariales para móviles.
  - Microsoft Azure App Service.
  - Google App Engine.

3. Service Models

- SaaS: Software as a Service
  - The capability provided to the consumer consists of using the provider's applications that run on a cloud infrastructure. The applications can be accessed from various client devices through a thin client interface, such as a web browser or a program interface.
  - The consumer does not manage or control the underlying cloud infrastructure, including the network, servers, operating systems, storage, or even the capabilities of individual applications, with the possible exception of user-specific application configuration settings.
  - Advantages:
    - Software that is constantly updated and common to all nodes.
    - Global service delivery.
  - Disadvantages:
    - Dependence on the external provider and on an Internet connection.
    - Security.
    - Long-term scalability.
  - Ejemplo: Microsoft Office 365, Google mail

3. Service Models

FaaS: Function as a Service (Serverless computing)

- Cloud-native applications.
- A cloud service model that allows users to create and run applications and processes without coming into contact with the underlying server.
- The required resources are provided on demand and in real time during the execution time of a single function call.
- Computing resources are managed by the Cloud provider
- Ventajas:
  - Agile resource provisioning.
  - Greater resource scalability
  - Costs are charged only for the resources that have actually been used.
- Inconvenientes:
  - The design and implementation of "serverless" structures require more effort.
  - Greater dependence on the provider.
- Ejemplo: Amazon AWS Lambda, MS Azure Functions, Google Cloud Functions

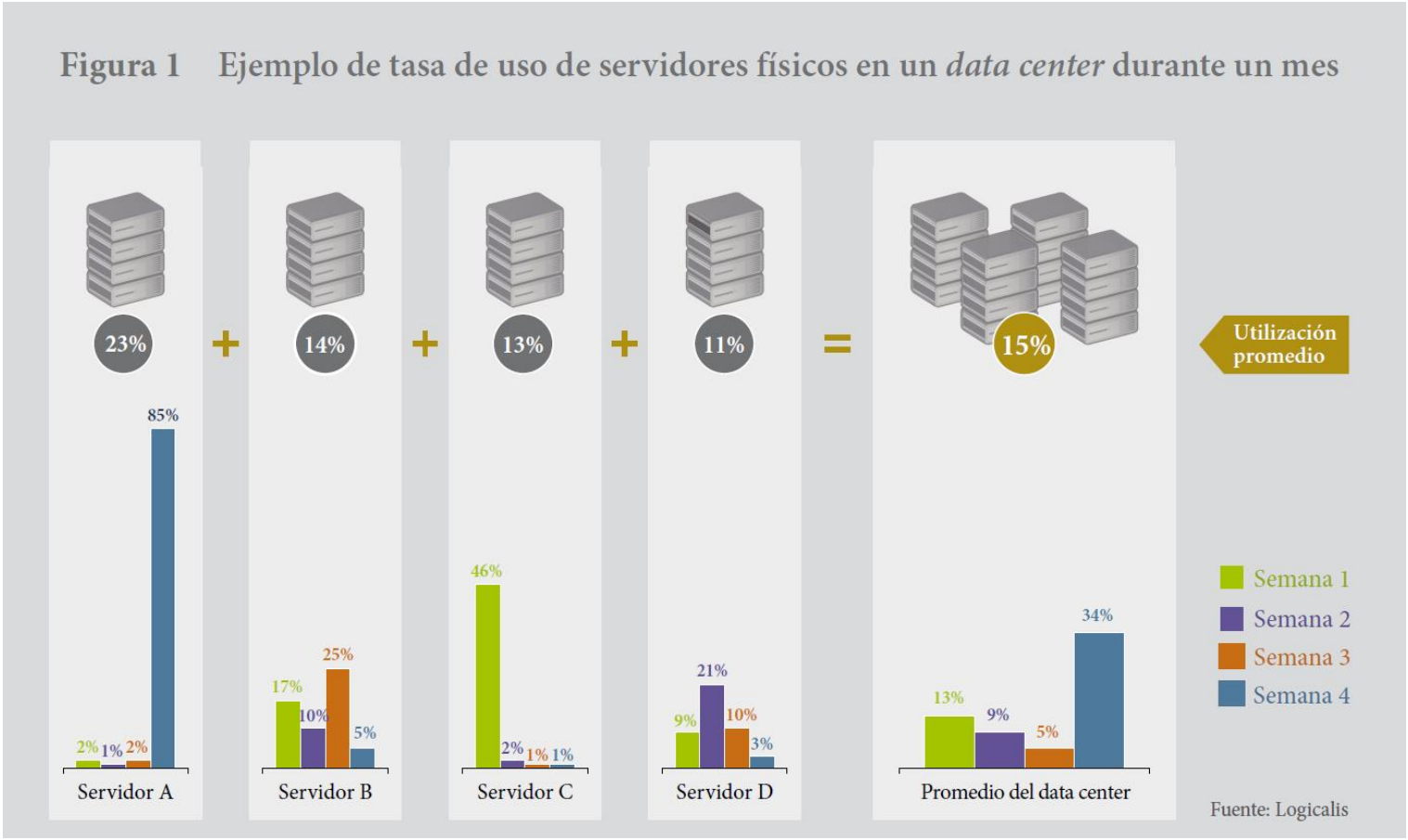
- Li Z. et al., (2022) The Serverless Computing Survey: A Technical Primer for Design Architecture, ACM Computing Surveys, Vol. 54, Issue 10, https://doi.org/10.1145/3508360

4. Virtualization

- Virtualization
  - Virtualization makes it possible to present a set of computing resources without configuration or location constraints.
  - Heterogeneity of machines and operating systems is allowed.
  - The improvement in server processing time makes it possible to reduce the number of machines required for the same workload.
  - Equipment and electrical energy costs are reduced.
  - It facilitates de centralization of hardware administration.

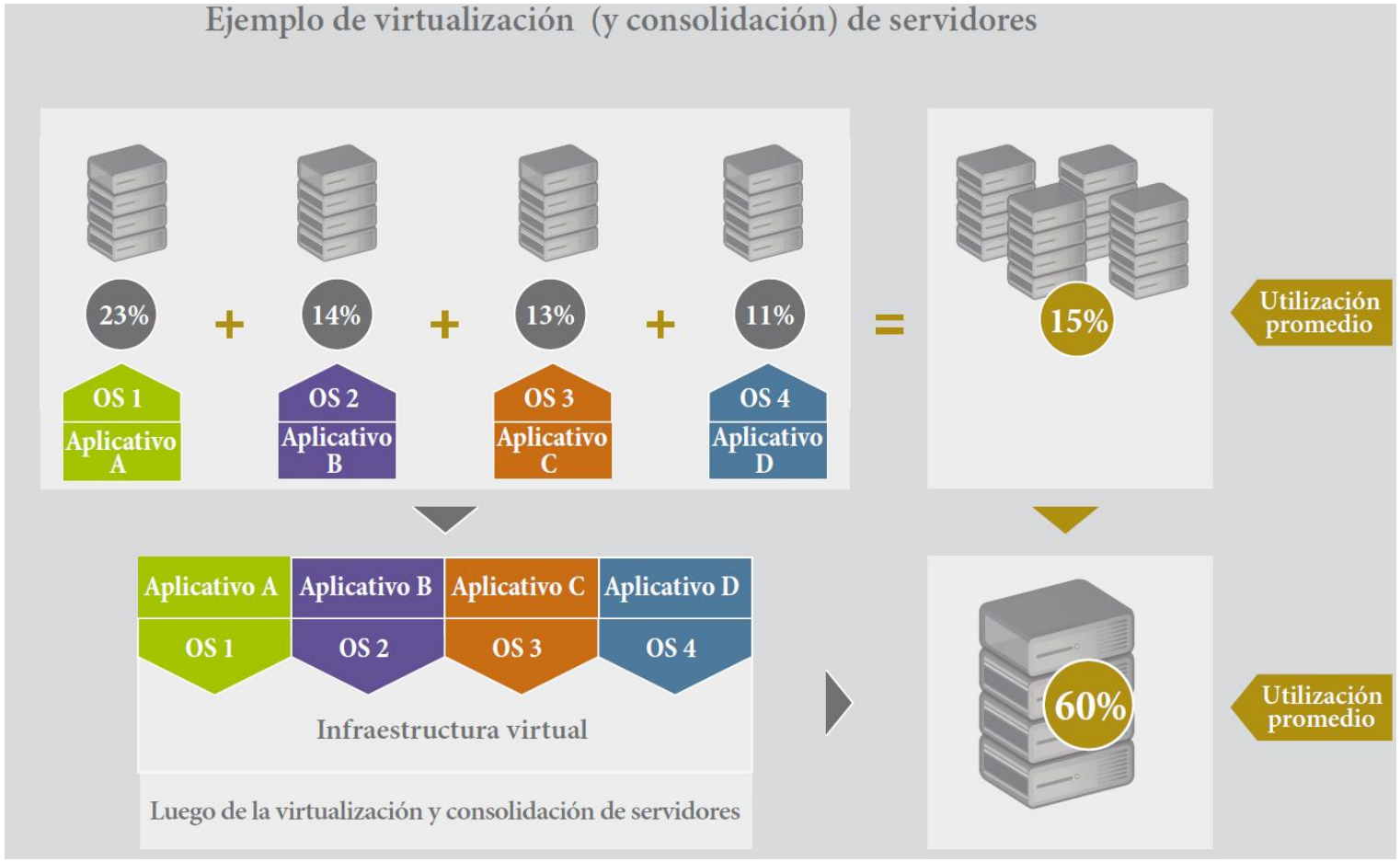
4. Virtualization

- Virtualization



4. Virtualization

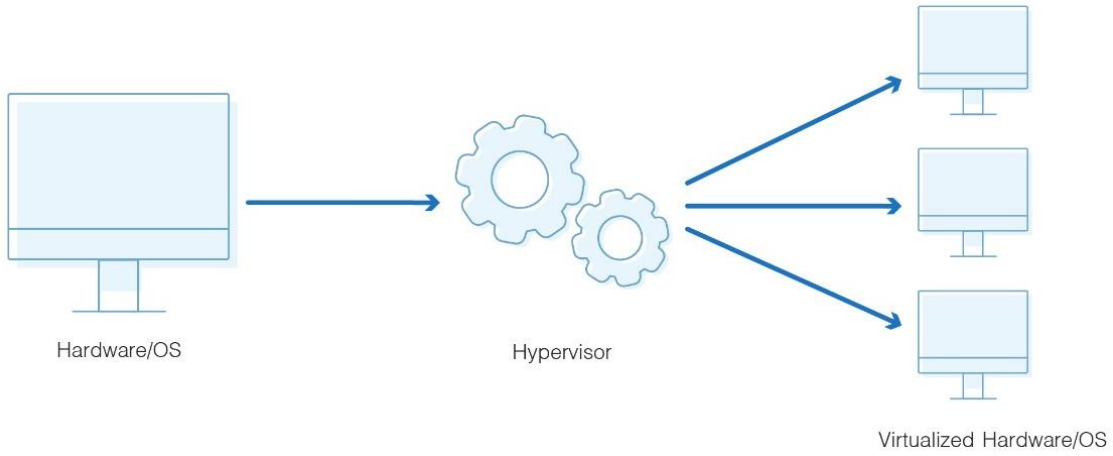
- Virtualization



4. Virtualization

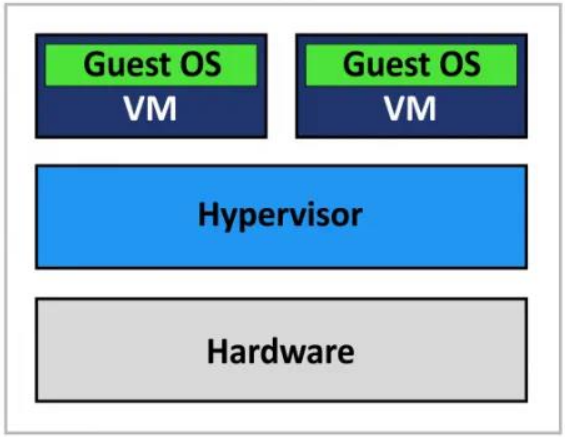
- Hypervisor

It is a system that creates and runs virtual machines through the virtual sharing of its resources.



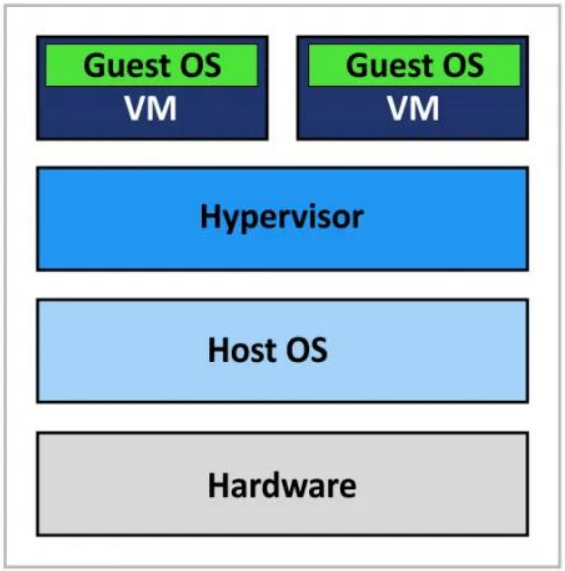
4. Virtualization

- Virtualization options



Type 1 Hypervisor
(Bare-Metal Architecture)

Ejemplos:
- Xen Server
- MS Hyper-V



Type 2 Hypervisor
(Hosted Architecture)

Ejemplos:
- VMware Workstation
- Oracle Virtual Box

5. Containers

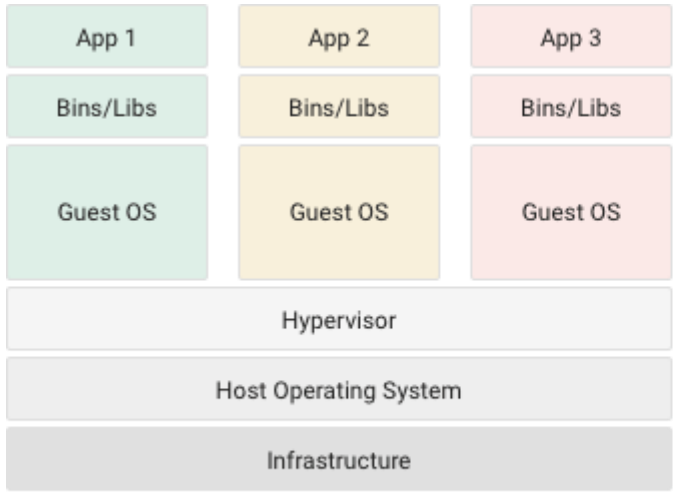
Containers

- A container is a standard software unit that packages code and all its dependencies so that the application runs quickly and reliably from one computing environment to another. A container image is a lightweight, standalone, executable software package that includes everything needed to run an application: code, runtime, system tools, system libraries, and settings. Container images become containers at runtime.
- In this way, containerized software will always run in the same manner, regardless of the infrastructure. Containers isolate software from its environment and ensure that it operates consistently despite differences, for example, between development and staging environments.

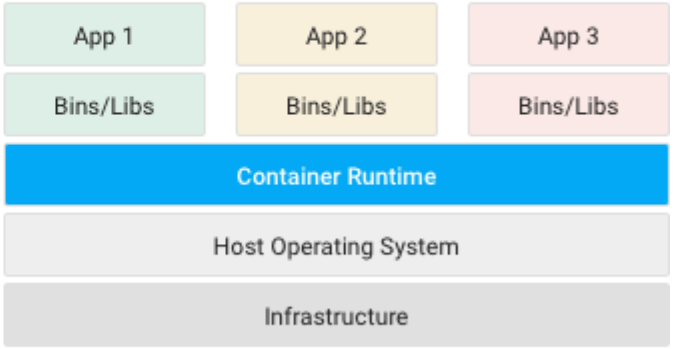
5. Containers

Contenedores

- Containers provide a much lighter solution than virtual machines, with which developers and IT operations teams can work.



Virtual Machines



Containers